# PROPER IS FINITE NEAR A FINITE FIBRE (RETAINED WITH THE AUTHOR'S SLOGAN BLOCK): FOR F : X -> S PROPER WITH S LOCALLY NOETHERIAN AND THE SET-THEORETIC FIBRE OVER S FINITE, SOME OPEN NEIGHBOURHOOD V OF S HAS F^{-1}(V) -> V FINITE.

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 6154-6172 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 6174-6188, the complete author proof (13 body lines).

*Editorial note.* Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Pool stacks-coherent backfill wave 33f.

*Editorial note.* Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-proper-finite-fibre-finite-in-neighbourhood' also exists in stacks-spaces-more-morphisms — a different proposition in a different chapter file.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

*Source reference.* Original lemma 02LS: sources/stacks-coherent/pinned-original-source.tex, source lines 6098–6107. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02UP: sources/stacks-coherent/pinned-original-source.tex, source lines 6188–6188. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02NG: sources/stacks-coherent/pinned-original-source.tex, source lines 6188–6188. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02NH: sources/stacks-coherent/pinned-original-source.tex, source lines 6188–6188. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

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*Date*: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 01TJ: sources/stacks-coherent/pinned-original-source.tex, source lines 6188–6188. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 01TI: sources/stacks-coherent/pinned-original-source.tex, source lines 6188–6188. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *(For a more general version see More on Morphisms, Lemma 02UP.) Let $f : X \to S$ be a morphism of schemes. Let $s \in S$. Assume*

(1) *$S$ is locally Noetherian,*
(2) *$f$ is proper, and*
(3) *$f^{-1}(\{s\})$ is a finite set.*

*Then there exists an open neighbourhood $V \subset S$ of $s$ such that $f|_{f^{-1}(V)} : f^{-1}(V) \to V$ is finite.*

*Proof.* The morphism $f$ is quasi-finite at all the points of $f^{-1}(\{s\})$ by Morphisms, Lemma 02NG. By Morphisms, Lemma 01TI the set of points at which $f$ is quasi-finite is an open $U \subset X$. Let $Z = X \setminus U$. Then $s \notin f(Z)$. Since $f$ is proper the set $f(Z) \subset S$ is closed. Choose any open neighbourhood $V \subset S$ of $s$ with $Z \cap V = \emptyset$. Then $f^{-1}(V) \to V$ is locally quasi-finite and proper. Hence it is quasi-finite (Morphisms, Lemma 01TJ), hence has finite fibres (Morphisms, Lemma 02NH), hence is finite by Lemma 02LS. □